

Bulgarian Solitaire: Recurrent Necklaces, Every Fixed Iterate, and the Full Finite Koopman Spectrum

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Abstract

For Bulgarian solitaire on every integer-partition set $\mathcal{P}(N)$, we turn Brandt's attributed recurrent classification into a complete periodic ledger. The unique decomposition $N = \binom{k}{2} + r$, $0 \leq r < k$, identifies recurrent partitions with length- k , weight- r binary words, and one solitaire move becomes rotation. We derive a closed formula for every positive-iterate fixed count, then recover all least periods, primitive cycles, and the finite Artin–Mazur zeta by Möbius inversion. The finite cycle factors also give the full Koopman algebraic spectrum, including the transient zero multiplicity. The finite census is a regression oracle, not the proof of the all- N theorem.

Keywords: Bulgarian solitaire; integer partitions; binary necklaces; primitive cycles; dynamical zeta; Koopman spectrum.

中文摘要

本文研究任意整数分拆集合上的保加利亚单人纸牌动力学。利用归属于 Brandt 的循环分拆分类，将循环状态与长度为 k 、重量为 r 的二进制词对应，其中 $N = \binom{k}{2} + r$ 且 $0 \leq r < k$ ；一次动力学迭代正好对应循环旋转。由此得到所有正迭代固定点数的闭式公式，再用 Möbius 反演恢复全部最小周期、原始循环、有限动力学 ζ 函数与包含瞬态零重数的完整有限 Koopman 代数谱。有限枚举仅用于回归检验，不替代全参数定理。

关键词：保加利亚单人纸牌；整数分拆；二进制项链；原始循环；动力学 ζ 函数；Koopman 谱。

1 The noninvertible map and Brandt coordinates

Write a partition of N as $\lambda = (\lambda_1 \geq \dots \geq \lambda_m > 0)$. The Bulgarian move is

$$T_N(\lambda) = \text{sort}(m, \lambda_1 - 1, \dots, \lambda_m - 1),$$

after zero parts are deleted. This map is generally noninjective. Every $N \geq 1$ has a unique representation

$$N = \binom{k}{2} + r, \quad k \geq 2, \quad 0 \leq r < k.$$

Brandt's cyclic-partition theorem [1], treated also by Akin and Davis [2], says that the recurrent set consists exactly of

$$\phi(w) = \text{positive parts of } (k-1, k-2, \dots, 0) + w, \quad w \in \{0, 1\}^k, \quad |w| = r.$$

For the right rotation $(\rho w)_i = w_{i-1 \bmod k}$, direct subtraction of the first column gives $T_N \phi = \phi \rho$. States correspond to words; dynamical cycles correspond to binary necklaces.

2 Every fixed iterate and every primitive cycle

Theorem 1 *Let $t \geq 1$ and $g = \gcd(k, t)$. Then*

$$F_t := \# \text{Fix}(T_N^t) = \begin{cases} \binom{g}{rg/k}, & (k/g) \mid r, \\ 0, & \text{otherwise.} \end{cases}$$

For each $d \mid k$, the number P_d of points of least period d and the number C_d of d -cycles are

$$P_d = \sum_{e \mid d} \mu(d/e) F_e, \quad C_d = P_d/d.$$

Consequently the zeta function of the full noninvertible finite map is

$$\zeta_{T_N}(z) = \exp \left(\sum_{t \geq 1} \frac{F_t}{t} z^t \right) = \prod_{d \mid k} (1 - z^d)^{-C_d}. \quad (1)$$

Proof. The index rotation ρ^t has g cycles, each of length k/g . A fixed binary word is constant on each index cycle, so its weight is a multiple of k/g ; choosing the rg/k one-cycles gives the displayed binomial. A point fixed by T_N^t is periodic and hence recurrent, so no transient partition adds to this count. Divisor Möbius inversion gives P_d , and division by d gives C_d . Substituting $F_t = \sum_{d \mid t} d C_d$ into the exponential definition proves (1). $\square$

3 The full finite Koopman algebraic spectrum

Let $p(N) = |\mathcal{P}(N)|$ and let $(U_N f)(\lambda) = f(T_N \lambda)$ on all complex functions on $\mathcal{P}(N)$. Ordering each functional-graph component from its transient trees toward its cycle gives

$$\det(\xi I - U_N) = \xi^{p(N) - \binom{k}{r}} \prod_{d|k} (\xi^d - 1)^{C_d}, \quad \det(I - zU_N) = \prod_{d|k} (1 - z^d)^{C_d} = \zeta_{T_N}(z)^{-1}. \quad (2)$$

Thus zero has algebraic multiplicity $p(N) - \binom{k}{r}$. If $\omega_k = e^{2\pi i/k}$, the nonzero multiplicities are

$$\text{mult}(\omega_k^j) = \sum_{\substack{d|k \\ k \nmid jd}} C_d, \quad \text{quad Tr}(U_N^t) = F_t.$$

Indeed, transient vertices contribute only a zero diagonal block to the characteristic polynomial, whereas a d -cycle contributes $\xi^d - 1$ and every d th root once. Formula (2) determines eigenvalues with algebraic multiplicities, not the nilpotent Jordan block sizes in the transient sector.

For $N = 8 = \binom{4}{2} + 2$, the six recurrent words form one 2-cycle and one 4-cycle. Hence $(F_1, F_2, F_3, F_4) = (0, 2, 0, 6)$, $\zeta_{T_8}(z) = ((1 - z^2)(1 - z^4))^{-1}$, zero has multiplicity $22 - 6 = 16$, and the four fourth-root multiplicities are $(2, 1, 2, 1)$. The sentinel is obtained both from all 22 partitions of 8 and from the word model.

4 Triangular boundary and evidence scope

If $r = 0$, the sole word is 0^k and the sole recurrent partition is the staircase $(k - 1, k - 2, \dots, 1)$. It is fixed, so $F_t = 1$ and $\zeta_{T_N}(z) = (1 - z)^{-1}$. The conclusion concerns the recurrent core; triangular deck size does not erase the transient basin in general.

The executable census covers every $1 \leq N \leq 40$. A producer enumerates 757 recurrent words and their 114 cycles. A separate checker constructs all 215,307 integer partitions, follows the actual noninvertible map, and recovers exactly the same recurrent states and cycles. This finite agreement tests implementations and conventions; it does not establish Brandt's all- N classification or classify complete transient trees and hitting times.

Scope. No target zero or prime table, arithmetic local datum, Euler factor, root number, automorphy input, target divisor, or Hilbert–Polya operator is used. The scope literal is `NO_BAD_EULER_OR_ROOT_NUMBER`.

References

- [1] J. Brandt, “Cycles of partitions,” *Proc. Amer. Math. Soc.* 85(3) (1982), 483–486. DOI: 10.1090/S0002-9939-1982-0656129-5.
- [2] E. Akin and M. Davis, “Bulgarian Solitaire,” *Amer. Math. Monthly* 92(4) (1985), 237–250. DOI: 10.1080/00029890.1985.11971590.